

Eric Rohr (he/him)

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Reserach Interests and Keywords

- galaxy formation and evolution, cosmic baryon cycle, multiphase gas
- galaxy clusters, the intracluster medium, circumgalactic medium
- galaxy environments, ram pressure stripping, jellyfish galaxies
- star-formation, stellar feedvack, supermassive black hole formation, growth, and feedback
- numerical methods, cosmological hydrodynamic simulations, idealized galaxy simulations

Education

Ph.D. Astronomy; Universität Heidelberg Oct 2020-Nov 2024
Jellyfish Galaxies and the Multiphase Nature of Gas Around Galaxies. Advisor: Dr. Annalisa Pilleich.

B.Sc. Astronomy-Physics; University of Virginia Aug 2016-May 2020
Why We Should Kerr About the Dark Secrets of Relativistic Accretion Disks in Athena++. Advisor: Prof. Shane Davis.

Academic Appointments

Postdoctoral Fellow June 2025-Present
Max Planck Institut für Astrophysik. Advisor: Volker Springel. Project: *The Effects of the Multiphase Halo Gas on Galaxy Evolution*.

Postdoctoral Researcher Dec 2024-May 2025
Max Planck Institut für Astronomie and STRUCTURES Cluster of Excellence. Advisor: Annalisa Pillepich. Project: *Zooming into the Tails of Jellyfish Galaxies*.

Ph.D. Student and IMPRS-HD Fellow Oct 2020-Nov 2024
Max Planck Institut für Astronomie. Advisor: Dr. Annalisa Pillepich. Thesis title: *Jellyfish Galaxies and the Multiphase Nature of Gas Around Galaxies*.

Publications

Full publication list on [ads](#). Selected recent works from 11 total peer-reviewed publications, including 4 as first author, 2 as second or third author, and 5 as contributing author.

- Staffehl, M., Nelson, D., Ayromlou, M. et al. incl. Rohr, E. (2025): “The abundance and origin of cool gas in galaxy clusters in the TNG-Cluster simulation”. [A&A 703, A33](#).
- Kurinchi-Vendhan, S., [Rohr, E.](#), Pillepich, A. et al. (2025): “Jellyfish galaxies with the IllustrisTNG simulations - Supermassive black hole activity in dense environments with ram-pressure stripped satellites”. [MNRAS 542, 1901](#).
- Rohr, E., Pillepich, A., Nelson, D., et al. (2025): “The cooler past of the intracluster medium in TNG-Cluster”. [MNRAS, 536, 1226](#).
- Rohr, [E.](#), Pillepich, A., Nelson D. et al. (2024): “The hot circumgalactic media of massive cluster satellites in the TNG-Cluster simulation: existence and detectability”. [A&A, 686, A86](#).

- Rohr, E., Pillepich, A., Nelson D. et al. (2023): “Jellyfish galaxies with the IllustrisTNG simulations - when, where, and for how long does ram pressure stripping of cold gas occur?”. [MNRAS](#), 524, 3502.

Conferences, Talks, and Seminars

Given 40 talks among conferences, seminars, and colloquia since 2020. Recent highlights:

- **Invited Talk** *Computational Galaxy Formation* conference: “Star-Formation in Mergers in the VESTRAL Simulations”. Ringberg Schloss, Germany. July 2026.
- **Invited Talk** *Confronting Simulations with the Real Universe* conference: “Comparing the Multiphase Circumgalactic Medium Across Cosmological Simulations”. Napa, California, USA. January 2026.
- **Contributed Talk** *Modeling of Multiphase Astrophysical Media* conference: “The Multiphase Circumgalactic Medium in Cosmological Simulation”. Schloss Ringberg, Germany. November 2025.
- **Colloquium** Max Planck Institute for Astronomy: “Origin and Fate of Cool Gas in and around Galaxy Clusters”. Heidelberg, Germany. April 2025.
- **Invited Talk** *Resolving the Circumgalactic Medium and Its Impact on Galaxy Evolution* conference: “Satellite Signatures in the Multiphase Halo Gas with the IllustrisTNG and TNG-Cluster Simulations”. Santa Cruz, Chile. November 2024.

Supervising

- Richards Albuquerque, FAPESP Fellow between MPA and University of São Paulo.
- Richa Shree, Master’s Thesis at MPA and University of Bochum. Now Ph.D. at Bochum.
- Shalini Kurinchi-Vendhan, Fulbright Fellow at MPIA. Now Ph.D. student at UCSD.
- Many students, high-school to Master’s level, through the MPA Mentorship Program.

Teaching

- Tutor for the Fortgeschrittenenpraktikum Wellenfrontanalyzse (Advanced Lab on Wavefront Analysis; FP36) at University of Heidelberg. Winter Semester 2022-23.
- Tutor for Cosmology (MVAstro4) at the University of Heidelberg. Summer 2021, 2022.
- TA for Observational Astronomy (ASTR3130) at the University of Virginia. Spring 2020.
- Tutor for AP Physics Physics as part of the Global Teaching Project. Fall 2019-Spring 2020.
- Instructor for the Philosophical Implications of Physics (INST1550) at the UVa. Spring 2019.

Service & Outreach

- Referee for MNRAS, A&A, and AAS Journals. 2022-Present.
- MPA Postdoc Representative Summer 2026-Present.
- Student Representative for the 16th generation of IMPRS-HD students. Fall 2020-Fall 2024.
- Volunteer at Explore Science public day in Mannheim, Germany (in German). June 2022.
- Student Rep. on the Graduate-Undergraduate Committee at UVa. Fall 2019-Spring 2020.
- Public Night volunteer at the UVa observatory. Fall 2017-Spring 2020.

Honors and Awards

- *D. Nelson Limber Prize* from the Department of Astronomy at University of Virginia in May 2020. **\$500.**
- *Alexander Vyssotsky Prize* from the Department of Astronomy at University of Virginia in May 2019. **\$1,000.**

- *Undergraduate Research Scholarship* from the Virginia Space Grant Consortium, a division of NASA, to be used at University of Virginia from August 2018-May 2019. **\$4,000.**
- *ThinkSwiss Research Scholarship* from the Office of Science, Technology, and Higher Education at the Embassy of Switzerland, to be taken at University of Zurich from May-August 2018. **4,800 CHF.**

Languages

 **Computer:** Python, C, C++, Fortran, html

 **Natural:** English (native), German (fluent, C1)

Academic References and Advisors

- Prof. Dr. Volker Springel, MPA ✉ vspringel@mpa-garching.mpg.de
- Dr. Annalisa Pillepich, MPIA ✉ pillepich@mpia.de
- Prof. Dr. Ralf Klessen, ZAH, ✉ klessen@uni-heidelberg.de
- Dr. Céline Péroux, ESO ✉ cperoux@eso.org